The documentation and process conversion measures necessary to comply with this revision shall be completed by 10 March 1998.

INCH POUND

MIL-PRF-19500/343D
10 December 1997
SUPERSEDING
MIL-PRF-19500/343C
18 June 1997

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, TRANSISTOR, NPN,
SILICON, SWITCHING, LOW POWER,
TYPES 2N2857 AND 2N2857UB
JAN, JANTX, JANTXV, AND JANS

This Specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for NPN, silicon, low power, ultra-high frequency transistors. Four levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figures 1 (similar to T0-72) and 3 (surface mount).

1.3 Maximum ratings.

P_T 1/ T_A = +25°C	P_T 2/ T_C = +25°C	I_C	V_{CBO}	V_{CEO}	V_{EBO}	T_J and T_{STG}
mW	mW	mA dc	V dc	V dc	V dc	°C
200	300	40	30	15	3	-65 to +200

1/ Derate linearly 1.14 mW/°C for TA > +25°C.
2/ Derate linearly 1.71 mW/°C for TC > +25°C.

1.4 Primary electrical characteristics at T_A = +25°C.

	h_{FE1}	h_{fe}	C_{cb}	F	G_{pe}	$r_b' C_c$
	V_{CE} = 1 V dc I_C = 3 mA dc	V_{CE} = 6 V dc I_C = 5 mA dc f = 100 MHz	V_{CB} = 10 V dc I_E = 0 f = 100 kHz ≤ f ≤ 1 MHz	V_{CE} = 6 V dc I_C = 1.5 mA dc f = 450 MHz R_g = 50 Ω (see figure 4)	V_{CE} = 6 V dc I_C = 1.5 mA dc f = 450 MHz	V_{CB} = 6 V dc I_E = 2 mA dc f = 31.9 MHz (see figure 5)
			pF	dB	dB	ps
Min	30	10	---	----	12.5	4
Max	150	21	1.0	4.5	21	15

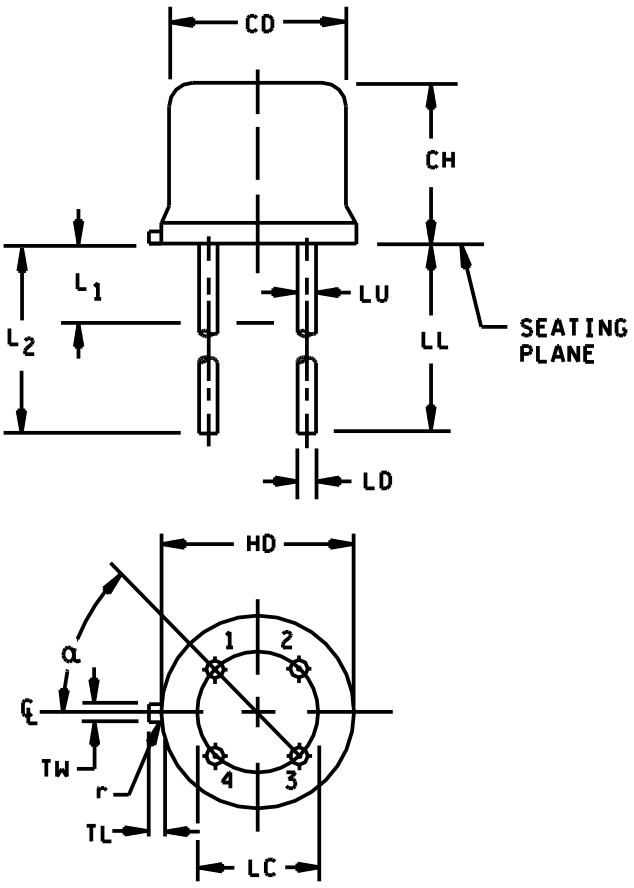
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad St., Columbus, OH 43216-5000, by using the addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

Distribution Statement A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:31Z
Check the source to verify that this is the current version before use.

Symbol	Inches		Millimeters		Note
	Min	Max	Min	Max	
CD	.178	.195	4.52	4.95	
CH	.170	.210	.8255	.8509	
HD	.209	.230	5.31	5.84	
LC	.100 TP		2.54 TP		6
LD	.016	.021	0.41	0.53	7,8
LL	.500	.750	12.70	19.05	7,8,13
LU	.016	.019	0.41	0.48	7,8
L_1	---	.050	---	1.27	7,8
L_2	.250	---	6.35	---	7,8
TL	.028	.048	0.71	1.22	3,4
TW	.036	.046	0.91	1.17	3
r	---	.010	---	0.25	10
α	45° TP		45° TP		6

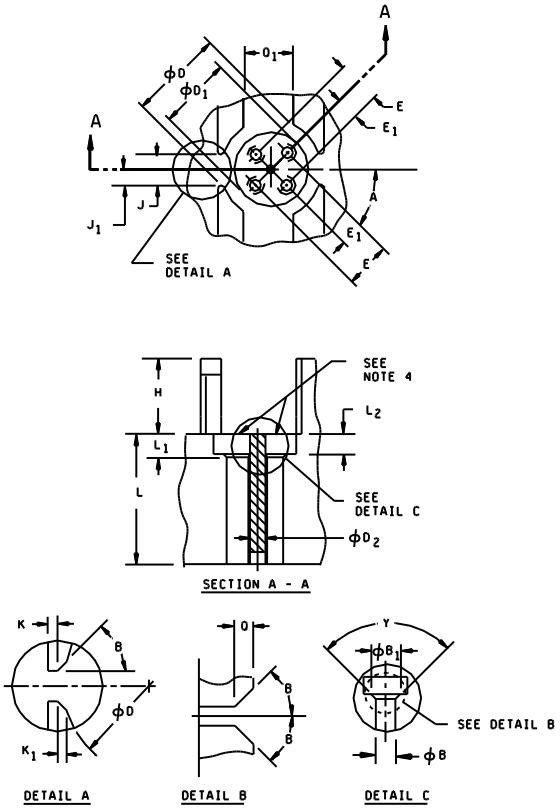


NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. Beyond r (radius) maximum, TH shall be held for a minimum length of .011 (0.28 mm).
4. Dimension TL measured from maximum HD.
5. Body contour optional within zone defined by HD, CD, and Q.
6. Leads at gauge plane .054 +.001 -.000 inch (1.37 +0.03 -0.00 mm) below seating plane shall be within .007 inch (0.18 mm) radius of true position (TP) at maximum material condition (MMC) relative to tab at MMC. The device may be measured by direct methods or by the gauge and gauging procedure shown in figure 2.
7. Dimension LU applies between L_1 and L_2. Dimension LD applies between L_2 and LL minimum. Diameter is uncontrolled in L_1 and beyond LL minimum.
8. All four leads.
9. The collector shall be internally connected to the case.
10. Dimension r (radius) applies to both inside corners of tab.
11. In accordance with ANSI Y14.5M, diameters are equivalent to ϕx symbology.
12. Lead 1 = emitter, lead 2 = base, lead 3 = collector.

FIGURE 1. Physical dimensions (similar to TO-72).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:31Z
Check the source to verify that this is the current version before use.

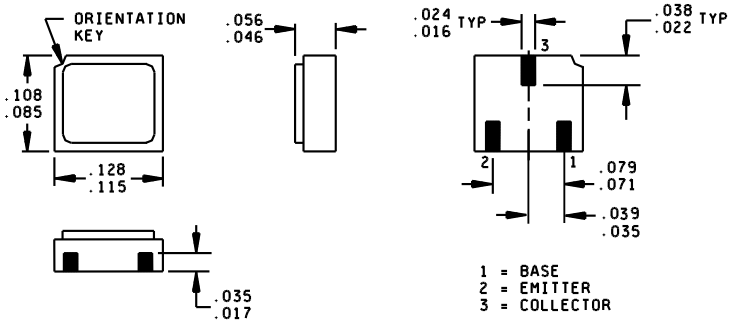
Ltr	Dimensions				Notes
	Millimeters		Inches		
	Min	Max	Min	Max	
ϕB	.8255	.8509	.0325	.0335	5
ϕB_1	1.09 Nom		.040 Nom		5
ϕD	5.867	5.880	.2310	.2315	
ϕD_1	4.04	4.09	.159	.161	
ϕD_2	1.02 Nom		.040 Nom		6
E	2.527	2.553	.0995	.1005	
E_1	1.257	1.283	.0495	.0505	
H	3.68	3.94	.145	.155	
J	1.194	1.207	.0470	.0475	
J_1	.597	.622	.0235	.0245	
K	.229	.279	.009	.011	
K_1	.127 Nom		.005 Nom		
L	9.45	9.60	.372	.378	
L_1	1.37	1.40	.054	.055	
L_2	1.09 Nom		.043 Nom		
Q	1.02 Nom		.040 Nom		
Q_1	3.12	3.23	.123	.127	
α	---	---	44.90°	45.10°	
B	---		45° Nom		
Y	---		90° Nom		



NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The following gauging procedures shall be used: The device being measured shall be inserted until its seating plane is .125 inch (3.18 mm) ±.010 inch (0.254 mm) from the seating surface of the gauge. A force of 8 ±.5 ounces shall then be applied parallel and symmetrical to the device's cylindrical axis. When examined visually after the force application (the force need not be removed) the seating plane of the device shall be seated against the gauge. The use of a pin straightener prior to insertion in the gauge is permissible. A spacer may be used to obtain the .125 inch (3.18 mm) distance from the gauge seat prior to force application.
4. These surfaces to be parallel and in same plane within ±.001 inch (0.025 mm).
5. Four holes.
6. Pressed in.

FIGURE 2. Gauge for lead and tab location.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:31Z
Check the source to verify that this is the current version before use.



Inches	mm	Inches	mm
.016	0.41	.056	1.42
.017	0.43	.071	1.81
.022	0.56	.079	2.01
.024	0.61	.085	2.41
.035	0.89	.108	2.74
.038	0.96	.115	2.82
.039	0.99	.128	3.25
.046	0.97		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 3. Physical dimensions, surface mount (UB version).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:31Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

C_{cb} - - - - - - - Collector to base feedback capacitance.

3.3 Interface requirements and physical dimensions. The Interface requirements and physical dimensions shall be as specified in MIL-PRF-19500, and figures 1 and 3 herein.

3.3.1 Lead finish. Lead finish shall be solderable as defined in MIL-PRF-19500.

3.4 Marking. Marking shall be in accordance with MIL-PRF-19500, except for the UB suffix package. Marking on the UB package shall consist of an abbreviated part number, the date code, and the manufacturers symbol or logo. The prefixes JAN, JANTX, JANTXV, and JANS can be abbreviated as J, JX, JV, and JS respectively. The "2N" prefix and the "UB" suffix can also be omitted.

3.5 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.2).

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in paragraph 1.3, 1.4, and table I.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in paragraphs 4.4.2 and 4.4.3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:31Z
Check the source to verify that this is the current version before use.

4. VERIFICATION

4.1 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500, and as specified herein.

4.3 Screening (JANS, JANTX, and JANTXV levels only). Screening shall be in accordance with MIL-PRF-19500 (table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV of MIL-PRF-19500)	Measurement	
	JANS level	JANTX and JANTXV levels
3c	Thermal impedance (see 4.3.2)	Thermal impedance (see 4.3.2)
7	hermetic seal (optional)	1/
9	I_{CBO1}, h_{FE2}	Not applicable
10	48 hours minimum	48 hours minimum
11	I_{CES}, h_{FE1}; ΔI_{CES} = 100% of initial value or 5 nA, whichever is greater; Δh_{FE1} = 15%;	I_{CES}, h_{FE1}
12	See 4.3.1	See 4.3.1
13	Subgroups 2 and 3 of table I herein; ΔI_{CES} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = 15%	Subgroup 2 of table I herein; ΔI_{CES} = 100% of initial value or 5 nA dc, whichever is greater; Δh_{FE1} = 15%

1/ Hermetic seal test shall be performed in either screen 7 or screen 14.

4.3.1 Power burn-in conditions. Power burn-in conditions are as follows:

V_{CB} = 15 Vdc, P_T = 200 mW at, T_A = Room ambient as defined in the general requirements of MIL-STD-750, paragraph 4.5;
NOTE: No heat sink or forced air cooling on the devices shall be permitted.

4.3.2 Thermal impedance ($Z_{\Theta JX}$ measurements). The $Z_{\Theta JX}$ measurements shall be performed in accordance with MIL-STD-750, Method 3131.

a. I_M measurement current - - - - - - - - - - - 10 mA.
b. I_H forward heating current - - - - - - - - 50 mA (min).
c. t_H heating time - - - - - - - - - - - - - - 20 - 30 ms.
d. t_{md} measurement delay time - - - - - - - - 60 µs max.
e. V_{CE} collector-emitter voltage - - - - - - - 10 V dc minimum

The maximum limit for $Z_{\Theta JX}$ under these test conditions are $Z_{\Theta JX}$ (max) = 55° C/W.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:31Z
Check the source to verify that this is the current version before use.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500, and as specified herein. If alternate screening is being performed per E.5.3.1d of MIL-PRF-19500, a sample of screened devices shall be submitted to and pass the requirements of group A1 and A2 inspection only (table VIb, group B, subgroup 1 is not required to be performed again if group B has already been satisfied per 4.4.2).

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-PRF-19500, and table I herein.

4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VIa (JANS) of MIL-PRF-19500 and 4.4.2.1. See 4.4.2.2 for JAN, JANTX, and JANTXV group B testing. Electrical measurements (end-points) and delta requirements shall be in accordance with the steps of table II herein as specified in the notes for table II.

4.4.2.1 Group B inspection, table VIa (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
B4	1037	V_{CB} = 10 V dc; P_T = 200 mW at T_A = room ambient per MIL-STD-750; t_{on} = t_{off} = 3 minutes minimum for 2,000 cycles. No heat sink or forced-air cooling on devices shall be permitted.
B5	1027	V_{CB} = 10 V dc; T_A = 125° C ± 25 ° C for 96 hours. P_T = 200 mW at +100° C or adjusted as required by the chosen T_A to give an average T_J = +275° C. Marking legibility requirements shall not apply.
B6	----	Not applicable.

4.4.2.2 Group B inspection, (JAN, JANTX, and JANTXV). 1/

Step	Method	Condition
1	1039	HTRB, test condition A, T_A = 150° C, V_{CB} = 24 V dc, 340 hours, n = 22 devices, c = 0.
2	1039	Steady-state life: Test condition B, 1,000 hours, V_{CB} = 15 V dc, P_T = 200 mW at T_A = room ambient per MIL-STD-750. No heat sink or forced-air cooling on the devices shall be
	or	permitted. n = 45 devices, Acc = 0
	1037	Intermittent operation life: V_{CB} = 15 V dc , 6,000 cycles, $\Delta T_J \geq$ +100° C; forced air cooling allowed on cooling cycle only. n = 45 devices, Acc = 0.
3	2037	Bond strength, n = 11 wires, c = 0.
4	1032	High-Temperature life (non-operating), T_A = +200° C n = 22, c = 0

1/ Separate samples may be used for each step. For rules on resubmission for failed steps, see MIL-PRF-19500 rules on resubmission of failed subgroups.

4.4.2.3 Group B sample selection. Samples selected from group B inspection shall meet all of the following requirements:

a. Samples shall be selected from each wafer lot from a die fabrication line that the manufacturer chooses to "certify" for the die design used in products manufactured to this specification.

b. Must be chosen from an inspection lot that has been submitted to and passed group A conformance inspection. When the final lead finish is solder or any plating prone to oxidation at high temperature, the samples for life test (subgroups B3 and B5 for JANS, and group B for JAN, JANTX, and JANTXV) may be pulled prior to the application of final lead finish.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:31Z
Check the source to verify that this is the current version before use.

4.4.2.4 Product acceptable for delivery. The product shall be acceptable for delivery only after the successful completion of all group B testing and shall be comprised of die meeting the following requirements:

a. Manufactured on the same certified die fabrication line as the sample selected for 4.4.2.3.

b. The die lot was manufactured (using start or completion dates, as the manufacturer selects to consistently use) within three months following the wafer lot used for sample selection in paragraph 4.4.2.3. Group B shall not be required more than once for any single wafer lot.

4.4.3 Group C inspection, JANS. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table VII of MIL-PRF-19500, and in 4.4.3.1 herein (JANS). See 4.4.3.2 for JAN, JANTX, and JANTXV group C testing. Electrical measurements (end points) and delta requirements shall be in accordance with the steps of table III herein as specified in the notes for table II.

4.4.3.1 Group C inspection, table VII (JANS) of MIL-PRF-19500.

Subgroup	Method	Condition
C2	2036	Test condition E; not applicable for UB devices.
C6	1026	conditions: 1,000 hours at V_{CB} = 15 V dc; P_T = 200 mW at T_A = room ambient per MIL-STD-750. No heat sink or forced-air cooling on device shall be permitted.

4.4.3.2 Group C inspection, JAN, JANTX, and JANTXV. Group C inspection (JAN, JANTX, and JANTXV, package related tests) shall be as follows and shall include those package or case-related tests which are performed periodically. Group C test on each package family shall be performed on devices from each six months production of devices (based on inspection lot identification codes) for each assembly location and die attach method. Electrical measurements (end points) and delta requirements shall be in accordance with the steps of table II herein as specified in the notes for table II. The sample size for each of these subgroups is: 22 devices, c = 0. Separate samples may be used for each step. For rules on resubmission for failed steps, see MIL-PRF-19500 rules on resubmission of failed subgroups.

Step	Method	Condition
1	2066	
2	1051	Test condition C.
3	2036	Test condition E; not applicable for UB devices.

4.4.3.3 Group C sample selection. Samples for subgroups in group C shall be chosen at random from any inspection lot containing the intended package type and lead finish procured to the same specification which is submitted to and passes group A tests for conformance inspection. Testing of a subgroup using a single device type enclosed in the intended package type shall be considered as complying with the requirements for that subgroup.

4.4.3.4 Product acceptable for delivery. The product which may be accepted for delivery based a successful completion of group C testing shall be product of the particular package type with lot inspection identification codes of the 52 successive weeks beginning with the inspection lot identification code of the successful group C sample (i.e. the inspection lot identification codes accepted for delivery based on successful group C testing will include those sealed during the approximate 10 weeks during which the sample was being screened and tested plus those sealed in the 42 weeks following successful completion of group C testing making a total of 52 successive calendar weeks.) A manufacturer shall not accept product for delivery until after successful completion of the group C testing on product of the applicable package type with an inspection lot identification code less than 51 weeks prior to the inspection identification code of the product being considered for acceptance.

4.5 Method of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

4.5.2 Noise figure. The noise figure shall be measured using commercially available test equipement and its associated standard test procedures (see figure 4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:31Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071	45 devices, c = 0				
Solderability 1/ 2/	2026	15 leads, c = 0				
Resistance to solvents 2/ 3/ 4/	1022	15 devices, c = 0				
Subgroup 2						
Breakdown voltage, collector to emitter	3011	Bias condition D; I_C = 3 mA dc;	$V_{(BR)CEO}$	15		V dc
Breakdown voltage, collector to base	3001	Bias condition D; I_C = 1 uA dc	$V_{(BR)CBO}$	30		V dc
Breakdown voltage, emitter to base	3026	Bias condition D; I_E = 10 uA dc	$V_{(BR)EBO}$	3		V dc
Collector to base cutoff Current	3036	Bias condition D; V_{CB} = 15 V dc	I_{CBO1}		10	nA dc
Forward-current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 3 mA dc	h_{FE1}	30	150	
Collector-emitter saturation voltage	3071	I_C = 10 mA dc; I_B = 1 mA dc;	$V_{CE(sat)}$		0.4	V dc
Base-emitter saturation voltage	3066	Test condition A; I_C = 10 mA dc; I_B = 1 mAdc;	$V_{BE(sat)}$		1.0	V dc
Collector to emitter cutoff Current	3041	Bias condition C; V_{CE} = 16 V dc	I_{CES}		100	nA dc
Subgroup 3						
High temperature operation		T_A = +150°C				
Collector to base cutoff current	3036	Bias condition D; V_{CB} = 15 V dc	I_{CBO2}		1.0	µA dc
Low temperature operation		T_A = -55°C				
Forward-current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 3 mA dc	h_{FE2}	10		

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:31Z
Check the source to verify that this is the current version before use.

TABLE I. Group A inspection

Inspection 1/	MIL-STD-750		Symbol	Limit		Unit
	Method	Conditions		Min	Max	
Subgroup 4						
Noise figure	----	V_{CE} = 6 V dc; I_C = 1.5 mA dc; f = 450 MHz R_g = 50 Ω (case lead grounded) (see 4.5.2 and fig. 4)	F		4.5	dB
Magnitude of common-emitter small-signal short-circuit forward-current transfer ratio	3306	V_{CE} = 6 V dc; I_C = 5 mA dc; f = 100 MHz (case lead grounded)	$\lvert h_{fe} \rvert$	10	21	
Small-signal short-circuit forward current transfer ratio	3206	V_{CE} = 6 V dc; I_C = 2 mA dc; (case lead floating)	h_{fe}	50	220	
Collector to base - feedback capacitance	----	V_{CB} = 10 V dc; I_E = 0 mA dc; 100 kHz ≤ f ≤ 1 MHz (see 4.5.1)	C_{cb}		1.0	pF
Small-signal power gain	3256	V_{CE} = 6 V dc; I_E = 1.5 mA dc; f = 450 MHz	G_{pe}	12.5	21	dB
Collector - base time constant	----	V_{CB} = 6 V dc; I_E = 2 mA dc; f = 31.9 MHz (see figure 5)	$r_b' C_c$	4	15	pF

1/ For sampling plan see MIL-PRF-19500.
2/ Separate samples may be used.
3/ Not required for laser marked devices.
4/ Not required for JANS devices.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:31Z
Check the source to verify that this is the current version before use.

TABLE II. Groups A, B, and C electrical end-point inspection measurements. 1/, 2/, 3/, 4/, 5/

Step	Inspection	MIL-STD-750		Symbol	Limit		Unit
		Method	Conditions		Min	Max	
1.	Collector-base cutoff current	3036	Bias condition D; V_{CB} = 15 V dc	I_{CBO1}		10	nA dc
2.	Collector - base cutoff current	3036	Bias condition D; V_{CB} = 15 V dc	I_{CBO2}		20	nA dc
3.	Collector to emitter voltage (saturated)	3071	I_C = 10 mA dc; I_B = 1.0 mA dc pulsed	$V_{CE(sat)}$		0.4	V dc
4.	Base-emitter voltage (saturated)	3066	Test condition A I_C = 10 mA dc; I_B = 1 mA dc	$V_{BE(sat)}$		1.0	V dc
5.	Forward current transfer ratio	3076	V_{CE} = 1 V dc; I_C = 3 mA dc	h_{FE1}	30	150	
6.	Collector-base cutoff current	3036	Bias condition D V_{CB} = 15 V dc	ΔI_{CB01}	100% of initial value or 5 nA dc, whichever is greater.		
7.	Forward current transfer ratio	3076	V_{CE} = 1.0 V dc; I_C = 3 mA dc;	Δh_{FE1}	±25% change from initial value.		
8.	Collector to emitter voltage (saturated)	3071	I_C = 10 mA dc; I_B = 1.0 mA dc	$\Delta V_{CE(sat)}$	± 50 mV dc change from initial value.		

1/ Devices which exceed the group A limits for this test shall not be accepted.

2/ The electrical measurements for table VIa (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 3, see table II herein, steps 1, 3, 4, 5, and 6.
   b. Subgroups 4 and 5, see table II herein, steps 2, 3, 4, 5, and 8.

3/ The electrical measurements for group B, 4.4.2.2 herein(JAN, JANTX, and JANTXV) are as follows:
   a. Steps 1, 2, and 4, of 4.4.2.2, see table II herein, steps 1, 6, 7, and 8.

4/ The electrical measurements for table VII (JANS) of MIL-PRF-19500 are as follows:
   a. Subgroup 2 and 3, see table II herein, steps 1, 2, 3, 4, 5, and 6 (for JANS).
   b. Subgroup 6, see table II herein, steps 2, 3, 4, 5, 6, 7, and 8 (for JANS).

5/ The electrical measurements for group C, 4.4.3.2 herein (JAN, JANTX, and JANTXV) are as follows:
   a. Step 2 of 4.4.3.2, see table II herein, all steps.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:31Z
Check the source to verify that this is the current version before use.

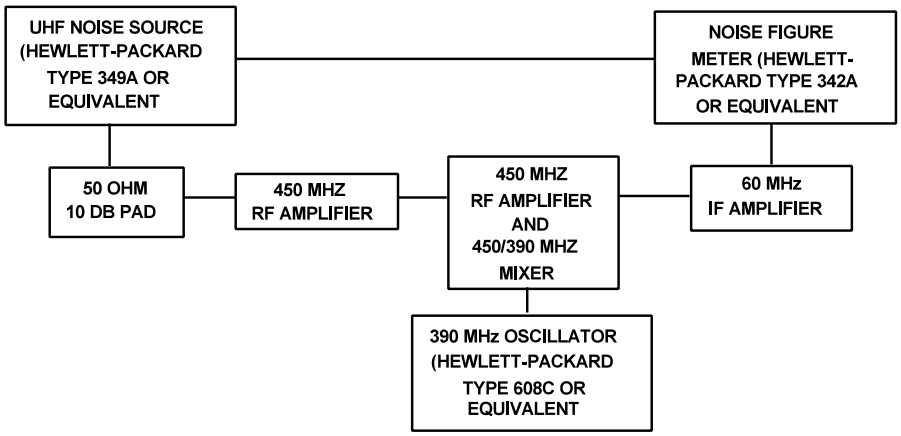
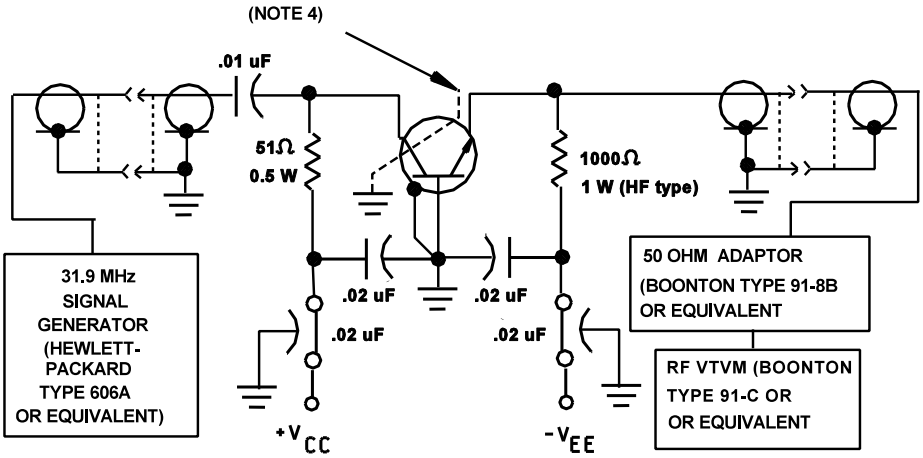


FIGURE 4. Block diagram for noise figure test.



Procedure:

1. With a short circuit applied between the collector and emitter terminals adjust 31.9 MHz input for 0.5 V RMS at emitter terminal of transistor.
2. After removing the short circuit between the collector and emitter circuit, insert unit to be tested and adjust V_{CC} and V_{EE} for V_{CB} = 6 V dc, I_E = 2 mA dc.
3. Read $r_b'C_C$ on RF voltmeter scale ($r_b'C_C$ in picoseconds = 10 times meter indication in millivolts) (1 millivolt = 10 picoseconds).
4. External interlead shield to isolate the collector lead from the emitter and base lead

FIGURE 5. Collector-base time constant test circuit (an equivalent circuit may be used).

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:31Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Lead formation and finish may be specified (see 3.3.1).

b. Type designation and product assurance level.

c. For die acquisition, the JANHC or JANKC letter version shall be specified (see figure 5).

d. Surface mount designation if applicable.

e. Packaging requirements (see 5.1).

6.3 Supersession information. Devices covered by this specification supersede the manufacturers' and users' Part or Identifying Number (PIN). The term Part or Identifying Number (PIN) is equivalent to the term part number which was previously used in this specification. This information in no way implies that manufacturers' PIN's are suitable as a substitute for the military PIN.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.5 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL-19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, DSCC-VQE, Columbus, OH 43216.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:31Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC
Air Force - 13, 19, 85, 99

Preparing activity:
DLA - CC

(Project 5961-1937)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:31Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	**1. DOCUMENT NUMBER** MIL-PRF-19500/343D	**2. DOCUMENT DATE (YYMMDD)** 97/12/10

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, TRANSISTOR, NPN, SILICON, SWITCHING, LOW POWER, TYPES 2N2857 AND 2N2857UB JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) DSN (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial DSN FAX E-mail 614-692-0510 850-0510 614-692-6939 alan_barone@dscc.dla.mil
c. ADDRESS (Include Zip Code) Defense Supply Center Columbus ATTN: DSCC-VAT Columbus, OH 43216-5000	**IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:** Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 DSN 289-2340

DD Form 1426, OCT 89 Previous editions are obsolete 198/290

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T14:31Z
Check the source to verify that this is the current version before use.